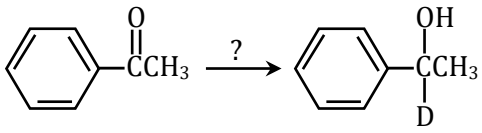
RACE
Regular Analysis Through Continuous Exercise

Alcohols and Ethers : RACE-01

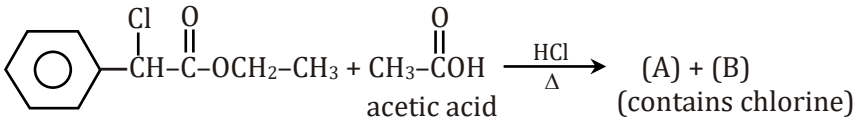
Preparation of Alcohols

1. Which of the following reagents would carry out the following transformation ? (D = ^2H)



(A) $NaBD_4$ in CH_3OH
(B) $LiAH_4$, then D_2O
(C) $NaBD_4$ in CH_3OD
(D) $LiAlD_4$, then D_2O

CAE188

2. 

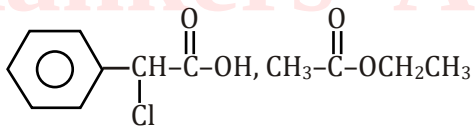
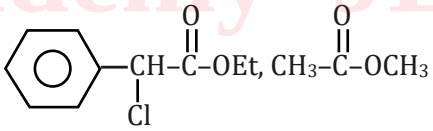
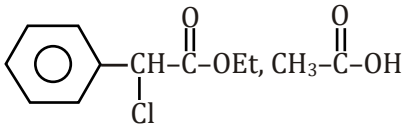
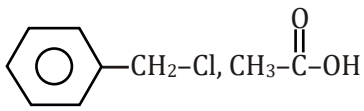
ethyl-2-chloro 2-phenyl acetate (1 mole)

acetic acid

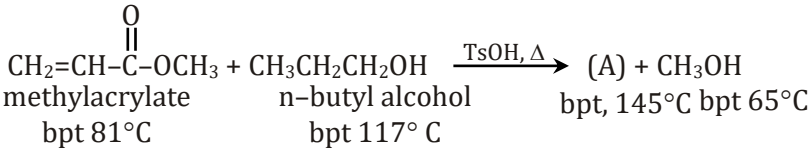
(A) + (B) (contains chlorine)

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Product (A) and (B) respectively in the above reaction are :

(A) 
(B) 
(C) 
(D) 

CAE189

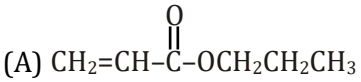
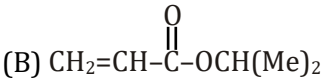
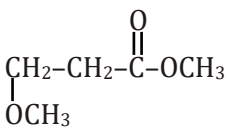
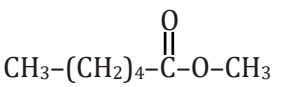
3. 

methylacrylate bpt 81°C

n–butyl alcohol bpt 117° C

(A) + CH_3OH bpt, 145°C bpt 65°C

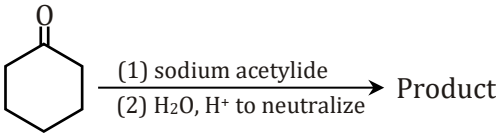
Product (A) in the above reaction are :

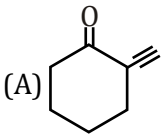
(A) 
(B) 
(C) 
(D) 

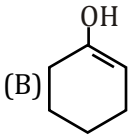
CAE190

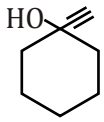
For More Join: @IITJEE_Advanced

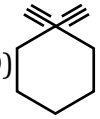
4. Give the major organic product of the following reaction.



(A) 

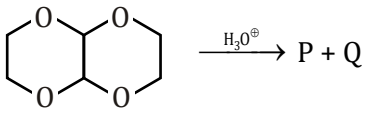
(B) 

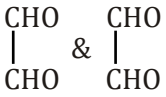
(C) 

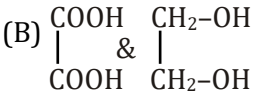
(D) 

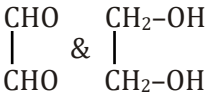
CAE191

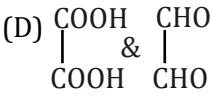
5. The acid catalysed hydrolysis products in the following reaction are -



(A) 

(B) 

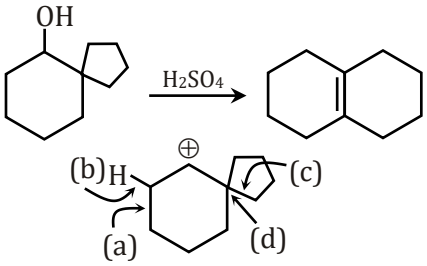
(C) 

(D) 

CAE195

Reaction of Alcohols and Ethers

6. The following transformation involves a carbocation rearrangement. The carbocation is generated by protonation of the hydroxyl group, followed by the loss of water. which bond has to migrate in the carbocation to yield the product indicated (after the deprotonation) ?

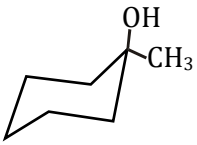


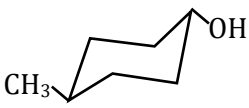
(A) a (B) b (C) c (D) d

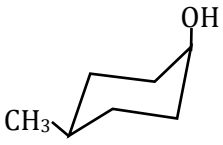
CAE176

7. Which of the following react with HBr at faster rate ?

(A) 

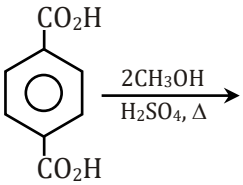
(B) 

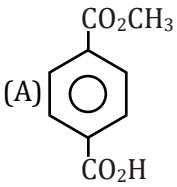
(C) 

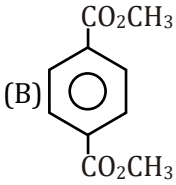
(D) 

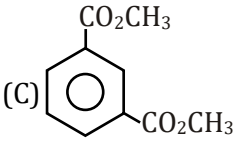
CAE177

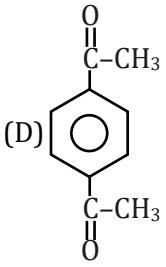
For More Join: @IITJEE_Advanced

8.  (A) ; Product (A) is:

(A) 

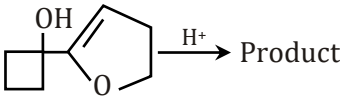
(B) 

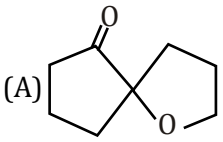
(C) 

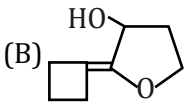
(D) 

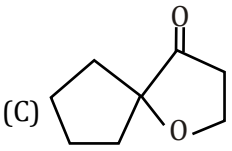
CAE178

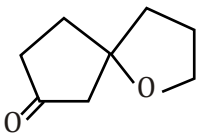
9. Identify the major product,



(A) 

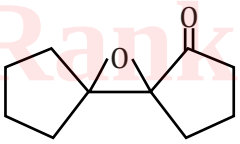
(B) 

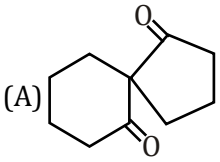
(C) 

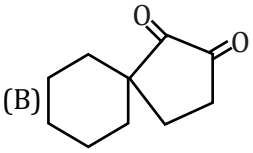
(D) 

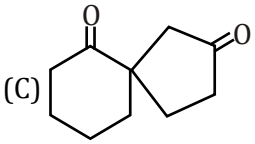
CAE179

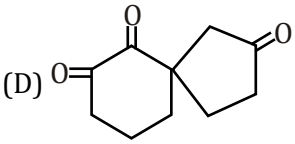
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10.  (A) ; Product (A) is

(A) 

(B) 

(C) 

(D) 

CAE180

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	A	C	C	C	B	B	A	A

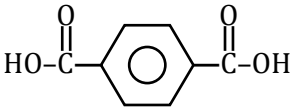
For More Join: @IITJEE_Advanced

Regular Analysis Through Continuous Exercise

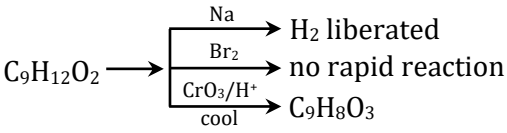
Alcohols and Ethers : RACE-02

Reaction of Alcohols and Ethers

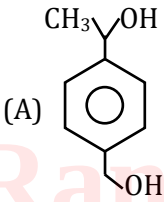
1. An optically active alcohol of formula $C_9H_{12}O_2$ produced the following compound when refluxed with $KMnO_4$.

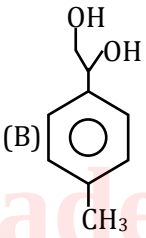


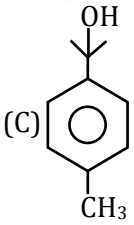
The original compound showed these properties also :



What is Structure of (A) ?







(D) both (A) and (B)

CAE186

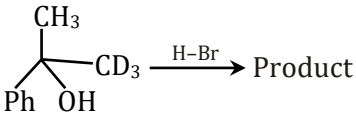
2. Which are not cleaved by HIO_4 ?

I : glycerol
II : glycol
III : 3–propenediol
IV : methoxy-2-propanol

(A) I,II,III,IV (B) I,II (C) II,III (D) III, IV

CAE187

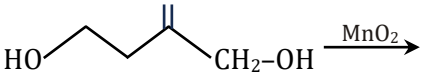
3. Which describes the best stereochemical aspects of the following reaction ?

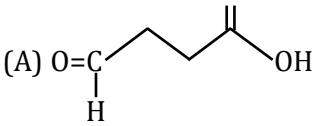


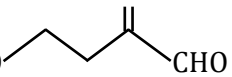
(A) Inversion of configuration occurs at the carbon undergoing substitution.
(B) Retention of configuration occurs at the carbon undergoing substitution.
(C) Racemization (loss of configuration) occurs at the carbon undergoing substitution.
(D) The carbon undergoing substitution is not stereo genic.

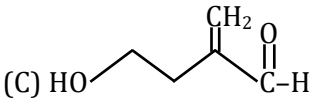
CAE192

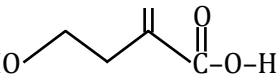
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For More Join: @IITJEE_Advanced

4.  (A) , Product (A) is :

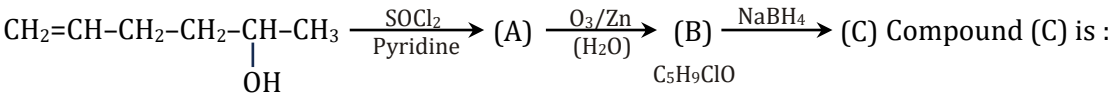
(A) 

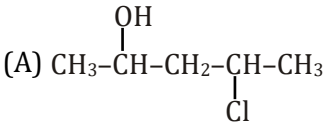
(B) 

(C) 

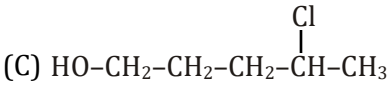
(D) 

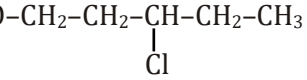
CAE193

5.  (C) Compound (C) is :

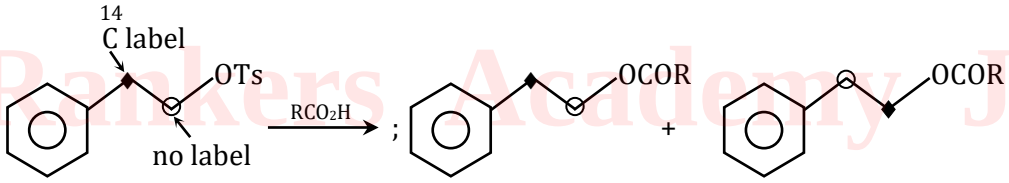
(A) 

(B) $HOCH_2-CH_2-CH_2-CH_2-Cl$

(C) 

(D) 

CAE194

6. 

(A) S_N1

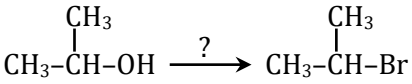
(B) S_N2

(C) SN–NGP

(D) SN–Ar

CAE181

7. Which is the best reagent to convert isopropyl alcohol to isopropyl bromide?



(A) HBr

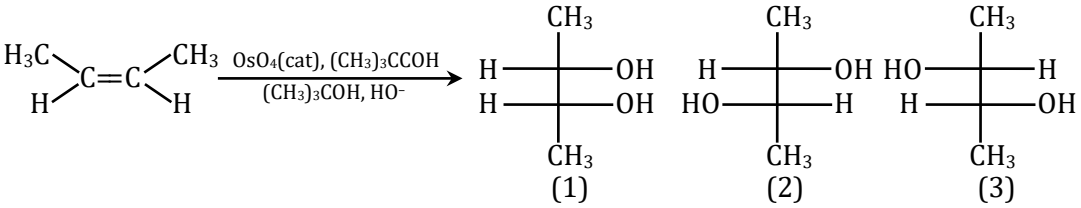
(B) $SOBr_2$

(C) Br_2

(D) CH_3MgBr

CAE182

8. What is the product of the following reaction?



(A) Only 1

(B) 1 : 1 mixture of 2 and 3

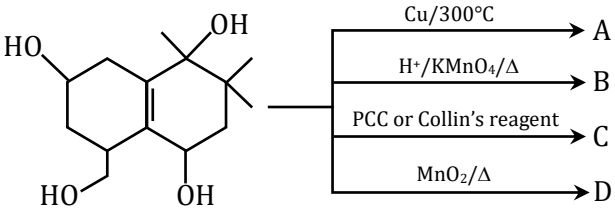
(C) Only 2

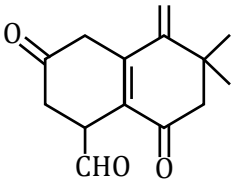
(D) 1 : 1 : 1 mixture of 1, 2 and 3

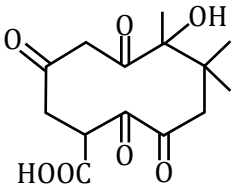
CAE183

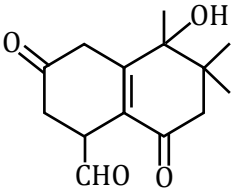
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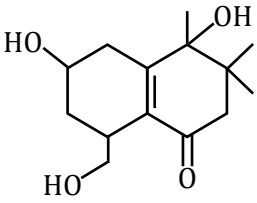
9. Complete the following reactions :



(A) 

(B) 

(C) 

(D) 

CAE185

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	D	D	C	C	C	C	B	A	ABCD

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